

April 4, 2026

1 Exploring Top Restaurants in Las Vegas: A Data Science Journey!

1.1 Background:

As we gear up for our summer trip to Las Vegas, we've decided to delve into the city's vibrant dining scene. To ensure we experience the best and most budget-friendly eateries, we've utilized the [Yelp dataset](#) to identify top-rated restaurants in Las Vegas. We've also filtered out the top 50 fast-food chains to avoid the usual options and embrace more unique culinary experiences.

1.1.1 Project Goal/Stakes:

Our objective is to analyze the Yelp dataset to pinpoint exceptional yet affordable dining establishments in Las Vegas. By focusing on local favorites and hidden gems, we aim to curate a list of restaurants that offer both quality and value, ensuring our trip is both delicious and economical.

1.1.2 Project Submission:

A Note from your Backpacking Partner:

"Hello, team. I've gathered and preprocessed the Yelp dataset for Las Vegas, filtering out the top 50 fast-food chains to ensure we explore the city's unique culinary offerings. Now, it's up to you to analyze this data and find the best budget-friendly restaurants for our trip. I request that you do NOT delete/modify any part of the question description, given code, and pre-loaded comments in the code block.

We look forward to your insightful analyses and recommendations for our culinary adventure in Las Vegas! Hope we can have a wonderful trip in Vegas."

Just a reminder, for a better a smooth autograder experience, please: - Write your answers between `# ANSWER STARTING HERE` and `# ANSWER ENDS HERE`. - Do not delete helper cells - Always use `seed=42` as your `random_state` in the whole notebook.

Submit this Google Colab/Jupyter Notebook file on Gradescope.

1.1.3 Part A (4 Pts Total)

You are expected to load the Yelp data and provide some metrics so I know what we are working with. Follow these steps.

1.1 Import Libraries (0 pt)

Many libraries are imported for you. Import the additional **Sci-kit learn** modules: - **Decision Tree Classifier** - **K Neighbors Classifier** - **Logistic Regression** - **Random Forest Classifier**

```
[46]: # Necessary libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split, cross_val_score, \
    ↳StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, \
    ↳confusion_matrix, silhouette_score

import matplotlib.pyplot as plt
import matplotlib
from sklearn.cluster import KMeans, DBSCAN
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
from sklearn.preprocessing import normalize

from matplotlib.colors import Normalize
from matplotlib.lines import Line2D

import ast
pd.set_option('display.max_columns', None)
from scipy.spatial.distance import cdist

import matplotlib.pyplot as plt
import seaborn as sns
import warnings # To suppress some warnings
import plotly.graph_objects as go
import plotly.express as px

# Suppress the specific FutureWarning
warnings.filterwarnings("ignore", category=FutureWarning, module="seaborn")

# Add the classifiers to the imports here:
# ANSWER STARTING HERE
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# ANSWER ENDS HERE
```

1.2 Understanding the Dataset (1 pt)

1. Load the dataset to a dataframe named `yelp_data`.
2. Print the dimensions of the dataset.
3. Print the column names of the dataframe.

```
[47]: # Load the yelp dataset
# ANSWER STARTING HERE
yelp_data = pd.read_csv('yelp_datasets.csv')
print(yelp_data.shape)
print(yelp_data.columns.tolist())

# ANSWER ENDS HERE
```

(21318, 16)

```
['business_id', 'name', 'city', 'state', 'latitude', 'longitude',
'business_stars', 'review_count', 'categories', 'checkins', 'review_id',
'user_id', 'reviewer_stars', 'useful', 'funny', 'cool']
```

```
[48]: yelp_data.head()
```

```
[48]:
```

	business_id	name	city	state	\
0	BjH8Xepc10i60hCDQdX6og	Emeril's New Orleans Fish House	Las Vegas	NV	
1	BCAw9a_kyx2RddRH9tMmKQ	Mr. Lucky's 24-7	Las Vegas	NV	
2	H8qpFitIesth86zqR4cwYg	Jean Philippe Patisserie	Las Vegas	NV	
3	K7lWdNUhCbcnEvIONhGewg	Wicked Spoon	Las Vegas	NV	
4	4mYP0zBokDgz307TW2EIKw	Crafted Buffet	Las Vegas	NV	

	latitude	longitude	business_stars	review_count	\
0	36.102795	-115.168508	3.5	943	
1	36.109100	-115.154526	3.5	802	
2	36.107727	-115.176986	4.0	1301	
3	36.109538	-115.176170	3.5	5950	
4	36.147248	-115.156032	2.5	159	

	categories	checkins	\
0	Seafood;American (New);Restaurants	13.0	
1	Restaurants;American (Traditional);Diners	7.0	
2	Specialty Food;Chocolatiers & Shops;Food;Desse...	62.0	
3	Buffets;Breakfast & Brunch;Restaurants	673.0	
4	Restaurants;Buffets	1.0	

	review_id	user_id	reviewer_stars	useful	\
0	DcYg8LW3RgeYZzLmWRODHw	4a0rhJSAJXFjLWGFZszfbQ	2.0	1.0	
1	3suTT5BSXAeyM3all0wOew	iB3iW2re2A4LQEFcpXExFw	3.0	0.0	
2	c2zN1d9eoNyV3AQshF7lsQ	GmQPriJGVlaHfpypcaT7SWg	3.0	1.0	
3	OMp-wasWFsrOo6MYX2zw6Q	IpvS0pJxjC6mNiK1MBSb4w	1.0	0.0	

4 w8PNyoNeQESah_TReX-VXA n9u_qpHgeHwarYbuux8B-w 3.0 2.0

	funny	cool
0	0.0	0
1	0.0	0
2	0.0	0
3	0.0	0
4	1.0	2

1.3 Explore the Data (3 pts)

Let's see what is stored in the dataset first to get a feel for what we can expect. Write only a **single line** of code for each. 1. Print the number of **samples** in the dataset (a numerical value). 2. Print the number of **features** in the dataset (a numerical value) 3. **Create two lists of the unique business names and categories**, named **names** and **categories**, respectively. (data type: **list**) Your answer should be similar to: - Business: ["Emeril's New Orleans Fish House", "Mr. Lucky's 24-7", 'Jean Philippe' ...] - Categories: ['Seafood', 'American (New)', 'Restaurants', 'American (Traditional)', ...]

```
[49]: # Print the number of samples
# ANSWER STARTING HERE
print("Number of samples:", yelp_data.shape[0])

# ANSWER ENDS HERE

# Print the number of features
# ANSWER STARTING HERE

print("Number of features:", yelp_data.shape[0])

# ANSWER ENDS HERE

# Write code to get your two lists.
# ANSWER STARTING HERE

# Extract and collect unique types
names = yelp_data['name'].unique().tolist()

categories = yelp_data['categories'].str.split(';').explode().unique().tolist()

# ANSWER ENDS HERE

# print("Business:", names)
# print(len(names))
# print("Categories:", categories)
# print(len(categories))
```

Number of samples: 21318
Number of features: 21318

1.1.4 Part B (9 Pts Total)

Now we need to do some scraping of the dataset so you can feed it into the training model.

2.1 Preprocessing (3 pts)

Before training the model, we need to clean and prepare the data. Please do the following. 1. Print the number of missing data in each column. Please specify which column name you got the number from. You can create a **dictionary** `missing_data`. 2. Impute missing values in the `yelp_data` dataset by filling them with zeros. Save the processed data as `yelp_data`.

```
[50]: # Print the number of missing data in each column.  
# ANSWER STARTING HERE  
missing_data = yelp_data.isnull().sum().to_dict()  
print(missing_data)  
  
# ANSWER ENDS HERE  
  
# ANSWER STARTING HERE  
  
yelp_data = yelp_data.fillna(0)  
  
# ANSWER ENDS HERE
```

```
{'business_id': 0, 'name': 0, 'city': 0, 'state': 0, 'latitude': 0, 'longitude':  
0, 'business_stars': 0, 'review_count': 0, 'categories': 0, 'checkins': 0,  
'review_id': 0, 'user_id': 0, 'reviewer_stars': 13, 'useful': 12, 'funny': 13,  
'cool': 0}
```

2.2 Splitting, Training, and Testing (6 pts)

Typically in machine learning, we wish to split the data into various portions in order to let a machine learning model train itself on one part of the data and test itself on another part of the data. Splitting also helps prevent overfitting, ensuring the model can make accurate predictions on new, unseen data.

Training data is used to learn the model, while testing data assesses its generalization to unseen examples. I would recommend using **80%** of the data for training and **20%** for testing.

Before we split the data, we are going to help you process the raw data a little bit more: 1. We prefer restaurants close to Canyon Gate in Las Vegas, near our Airbnb. 2. We add a `preference_rating`

column at the end of the dataframe, so that we can predict the rating classification. 3. We add non-linearity to `preference_rating`

```
[51]: # Helper Function, Do Not Remove
import numpy as np
import pandas as pd
import hashlib
from sklearn.preprocessing import MinMaxScaler

def business_id_to_float(business_id):
    hash_object = hashlib.md5(business_id.encode())
    hash_hex = hash_object.hexdigest()
    hash_int = int(hash_hex[:8], 16)
    return hash_int / (2**32 - 1)

def float_to_business_id(unique_float, original_business_ids):
    id_to_float = {id: business_id_to_float(id) for id in original_business_ids}
    closest_id = min(id_to_float.keys(), key=lambda x: abs(id_to_float[x] -
↵unique_float))
    return closest_id

# Non-linear transformation functions
def apply_non_linear_transformation(x, steepness=2):
    """
    Apply a sigmoid-like non-linear transformation
    - Increases sensitivity around mid-range values
    - Compresses extreme values
    """
    return 1 / (1 + np.exp(-steepness * (x - 0.5))) * 8

def exponential_boost(x, base=1.5):
    """
    Apply exponential boosting to favor higher values
    """
    return np.power(x, base)

def angular_score_adjustment(x):
    """
    Use trigonometric function to add non-linear variation
    """
    return np.sin(np.pi * x / 1.5) * 2

# Define weights
weights = {
    'location_score': 0.15,
    'business_stars': 0.35,
    'reviewer_stars': 0.35,
```

```

        'engagement_score': 0.15
    }

# Define the target location as Canyon Gate, Las Vegas
target_location = (36.1433, -115.2821)

# Haversine distance calculation
def haversine(lat1, lon1, lat2, lon2):
    R = 6371 # Earth radius in kilometers
    phi1, phi2 = np.radians(lat1), np.radians(lat2)
    dphi = np.radians(lat2 - lat1)
    dlamba = np.radians(lon2 - lon1)
    a = np.sin(dphi / 2)**2 + np.cos(phi1) * np.cos(phi2) * np.sin(dlamba / 2)**2
    return 2 * R * np.arctan2(np.sqrt(a), np.sqrt(1 - a))

# Main processing function
def generate_preference_rating(yelp_data):
    # Calculate distance from target location
    yelp_data['distance_km'] = yelp_data.apply(
        lambda row: haversine(target_location[0], target_location[1],
                               row['latitude'], row['longitude']), axis=1
    )

    # Normalize distance to [0, 1] and invert (closer is better)
    scaler = MinMaxScaler()
    yelp_data['location_score'] = 1 - scaler.
    fit_transform(yelp_data[['distance_km']])

    # Normalize business_stars and reviewer_stars to [0, 1]
    yelp_data['business_stars_norm'] = yelp_data['business_stars'] / 5
    yelp_data['reviewer_stars_norm'] = yelp_data['reviewer_stars'] / 5

    # Calculate engagement_score and normalize
    yelp_data['engagement_score_raw'] = yelp_data['useful'] +
    yelp_data['funny'] + yelp_data['cool']
    yelp_data['engagement_score'] = scaler.
    fit_transform(yelp_data[['engagement_score_raw']])

    # Calculate preference_rating with non-linear transformations
    yelp_data['preference_rating'] = (
        weights['location_score'] *
    apply_non_linear_transformation(yelp_data['location_score']) +
        weights['business_stars'] *
    exponential_boost(yelp_data['business_stars_norm']) +

```

```

        weights['reviewer_stars'] *
    ↪ angular_score_adjustment(yelp_data['reviewer_stars_norm']) +
        weights['engagement_score'] * np.log1p(yelp_data['engagement_score'])
    )

    # Normalize the preference_rating to 1-10 scale
    rating_scaler = MinMaxScaler(feature_range=(1, 10))
    yelp_data['preference_rating'] = rating_scaler.fit_transform(
        yelp_data[['preference_rating']]
    ).round().astype(int)

    # Convert business_id to float
    yelp_data['business_id'] = yelp_data['business_id'].
    ↪ apply(business_id_to_float)

    return yelp_data

# Data preparation and display
def prepare_final_ranking(yelp_data):
    # Remove duplicates, keeping highest-rated business for each name
    unique_data = yelp_data.sort_values(by='preference_rating',
    ↪ ascending=False).drop_duplicates(subset='name', keep='first')

    # Create ranked dataframe
    ranked_yelp = unique_data[['name', 'preference_rating']].
    ↪ sort_values(by='preference_rating', ascending=False)

    return ranked_yelp

# Process the data
yelp_data = generate_preference_rating(yelp_data)

# Prepare and display ranking
ranked_results = prepare_final_ranking(yelp_data)
print(ranked_results)

```

	name	preference_rating
15475	Eatt Gourmet Bistro	10
2420	La Maison de Maggie	10
1937	Papa Pinny's Pizzeria	10
17725	Pan Asian	10
18171	Patisserie Manon	10
...	...	...
15806	The Lodge at Shelbourne	3
14441	Magura	3
11240	The Lodge Tenaya	3
9418	Rosati's	3

[1474 rows x 2 columns]

Using every feature as input to the training process is not a good idea. Some features are not related to business rating, so let's exclude them from our feature set. Let the criteria for a good restaurant be a rating (preference_rating) greater than **eight**.

1. Extract the following for the input data X: business_stars, review_count, checkins, reviewer_stars, useful and cool.
2. Store this binary classification variable in Y.

```
[52]: # Initialize X and Y
# ANSWER STARTING HERE
X = yelp_data[['business_stars', 'review_count', 'checkins', 'reviewer_stars',
↪ "useful", "funny", "cool"]]
Y = (yelp_data['preference_rating'] > 8).astype(int)

print(X.head())
print(Y.head())

# ANSWER ENDS HERE
```

	business_stars	review_count	checkins	reviewer_stars	useful	funny	cool
0	3.5	943	13.0	2.0	1.0	0.0	0
1	3.5	802	7.0	3.0	0.0	0.0	0
2	4.0	1301	62.0	3.0	1.0	0.0	0
3	3.5	5950	673.0	1.0	0.0	0.0	0
4	2.5	159	1.0	3.0	2.0	1.0	2

0	0
1	0
2	0
3	0
4	0

Name: preference_rating, dtype: int64

3. Use `train_test_split` from the Scikit-learn module to split the training data using the proposed 80/20 ratio.

```
[53]: seed = 42 # always use this seed number as your random_state in the whole
↪ notebook
test_size=0.2

# Split your training data and use the seed to get consistent results.
# ANSWER STARTING HERE

X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=test_size,
↪ random_state=seed)
```

```
# ANSWER ENDS HERE
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(17054, 7)
(4264, 7)
(17054,)
(4264,)
```

4. **Normalize** both the training and test sets so that the model is trained on features with a consistent scale. **You can use any scaler from the Scikit-learn module to do this.** If we don't normalize the data, the model may struggle to converge during training because some features could dominate the loss function due to their larger scale. By normalizing the data, we ensure that each feature contributes equally to the learning process, improving the model's ability to find an optimal solution. (eg. [StandardScaler documentation](#))

```
[54]: #You can add more than 1 scalar. This is the base code.
# ANSWER STARTING HERE

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
# ANSWER ENDS HERE
```

1.1.5 Part C (15 Pts Total)

This is going to be the main source of evaluation for the return offer. Be thorough with your model search. You will train a Logistic Regression Model, plus 3 different classifiers: KNN, DecisionTree, Logistic Regression (with maximum 1000 iterations) and Random Forest.

3.1 Load Models (4 pts)

1. Create a python dictionary named `models` that maps the names of the models to their constructors.

```
[55]: # Initialize the models
# ANSWER STARTING HERE
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=seed),
    "Decision Tree": DecisionTreeClassifier(random_state=seed),
    "KNN": KNeighborsClassifier(),
    "Random Forest": RandomForestClassifier(random_state=seed)
}
```

```
# ANSWER ENDS HERE
```

3.2 K-fold Cross-Validation (8 pts) Using the standardized feature sets, do the following:

1. Given five folds and the object with the specified number of splits, perform **k-folds cross-validation** for **each** model using StratifiedKFold.
2. Print the **name** of the model and **cross validation accuracy** (mean and standard deviation) for each.

[Cross Validation Documentation](#)

```
[56]: # Perform k-fold Cross-Validation for each model
k_folds = 5
skf = StratifiedKFold(n_splits=k_folds, shuffle=True, random_state=seed)

for model_name, model in models.items():
    np.random.seed(42)
    print(f"Model: {model_name}")
    # ANSWER STARTING HERE

    cv_scores = cross_val_score(model, X_train_scaled, y_train, cv=skf)
    mean_score = cv_scores.mean()
    std_score = cv_scores.std()

    print(f"Cross-validated Accuracy: {mean_score} (+/- {std_score})")
    # ANSWER ENDS HERE
```

Model: Logistic Regression

Cross-validated Accuracy: 0.9329189761260576 (+/- 0.0008765375142048607)

Model: Decision Tree

Cross-validated Accuracy: 0.9642900706786994 (+/- 0.002583075206596999)

Model: KNN

Cross-validated Accuracy: 0.9299283412041944 (+/- 0.0021504078052851346)

Model: Random Forest

Cross-validated Accuracy: 0.9539112978452498 (+/- 0.002575361840192982)

3.3 Question: What is the most accurate ML model from above choice? Why? (3 pts) Based on the results above, the Decision Tree is the most accurate model with an accuracy of around 96.43%. While the others aren't far behind, this one seems to best predict the most important factors that lead to a good restaurant (preference_rating > 8). Even though it has the highest standard deviation of the models, they are all extremely low, which indicates that model is not guessing and using a structured formula to predict top tier restaurants.

1.1.6 Part D (12 Pts Total)

We want you to deploy of all the models to production. Evaluate all of the models on the heldout test set.

4.1 Model Evaluation (6 pts)

For each of the models do the following in order to evaluate each of them properly: - **Train** each of the models - Print each of the **accuracies** (3 decimal precision) of the models by their **names** in this format: *Accuracy of (model name): (accuracy of the model)*. - Print a **classification report** with the accuracy, precision, recall and f1-score of each model.

Note: The classes in your target variable are currently labeled as True and False. In the classification report, these represent the actual class labels (e.g., True = Good Rating, False = Bad Rating), NOT whether the model's prediction was correct or incorrect.

```
[57]: # Train each model using the training data
for model_name, model in models.items():
    # Set random seed for reproducibility
    np.random.seed(42)

    # ANSWER STARTING HERE
    model.fit(X_train_scaled, y_train)

    # ANSWER ENDS HERE

# Evaluate the performance of each model
for model_name, model in models.items():
    np.random.seed(42)

    # ANSWER STARTING HERE
    y_pred = model.predict(X_test_scaled)
    accuracy = accuracy_score(y_test, y_pred)
    print(f"Accuracy of {model_name}: {accuracy:.3f}") # Your accuracy table
    ↳header here
    print(classification_report(y_test , y_pred, target_names=['Bad Rating',
    ↳(False)', 'Good Rating (True)']))
    # ANSWER ENDS HERE
```

Accuracy of Logistic Regression: 0.931

	precision	recall	f1-score	support
Bad Rating (False)	0.93	1.00	0.96	3974
Good Rating (True)	0.36	0.01	0.03	290
accuracy			0.93	4264
macro avg	0.65	0.51	0.50	4264
weighted avg	0.89	0.93	0.90	4264

Accuracy of Decision Tree: 0.964

	precision	recall	f1-score	support
Bad Rating (False)	0.98	0.98	0.98	3974
Good Rating (True)	0.73	0.74	0.74	290

accuracy			0.96	4264
macro avg	0.86	0.86	0.86	4264
weighted avg	0.96	0.96	0.96	4264

Accuracy of KNN: 0.924

	precision	recall	f1-score	support
Bad Rating (False)	0.95	0.97	0.96	3974
Good Rating (True)	0.40	0.24	0.30	290

accuracy			0.92	4264
macro avg	0.67	0.61	0.63	4264
weighted avg	0.91	0.92	0.91	4264

Accuracy of Random Forest: 0.949

	precision	recall	f1-score	support
Bad Rating (False)	0.96	0.98	0.97	3974
Good Rating (True)	0.69	0.47	0.56	290

accuracy			0.95	4264
macro avg	0.83	0.73	0.76	4264
weighted avg	0.94	0.95	0.94	4264

4.2 Model Interpretation (6 pts)

-
- **Interpret** the results of the classification models (list the models which got the best scores for Accuracy, Precision, Recall and F1 score).
 - Clarification: When discussing Precision, Recall, and F1 score, focus your analysis specifically on the model's ability to predict the positive class ("True" / Good Rating), as identifying good restaurants is our primary objective.
 - **Recommend the best model according to your analysis.** Hopefully your best model has over a 90% accuracy as this will guarantee your return offer.

Question: Which model according to your analysis was the best at predicting the Restaurant Preference Rating? Why? (Focus on the metrics for the "True" class in your justification. DO NOT just say that the accuracy is better than the rest of the models. Dive into what precision and recall mean for this specific scenario.)

Based on the results above, the Decision Tree was the best at predicting restaurant preference rating. First of all, the accuracy of this model was higher than any of the other models. But, the most important reason why it is the best is because of the recall values. Recall refers to how well the model can point out all the good/bad spots in the city. The recall for the bad rating restaurants are all really high which signals that they all identify nearly all the bad restaurants as bad. However, only the Decision Tree is over 50% for recall for the good rating restaurants at 74%. This means that it still has a pretty good rating in identifying good restaurants and not just

guessing everything is a bad restaurant (looks like what Logistic Regression was doing). Another factor is the precision. Precision is how often the model was right when claiming a restaurant was good. The precision of the Decision Tree model is the highest, indicating that when it makes a claim, it is almost always right.

1.1.7 Part E (5 Pts Total)

Seeing all the effort and hard work, it looks as if most people have done well with this project! With all the input features you have used, there is still no concrete way of figuring out whether a feature is more important than another. We only really used intuition for that. Thus, we shall do some feature engineering to determine this.

5.1 What are your top 4 features used to predict Our restaurant preference rating (5 pts)?

- Use Decision Tree to create a dataframe for **feature importance** named **feature_importance_df**.
- **Extract, Sort, and Print** the **top 4** features from that dataframe.

Here are some references to help you:

[Feature Importance 1](#)

[Feature Importance 2](#)

```
[58]: # Extract the feature importances from the most accurate model
# ANSWER STARTING HERE
importances = models["Decision Tree"].feature_importances_

# Creating a DataFrame to display feature importances
feature_importance_df = pd.DataFrame({'Feature': X.columns, 'Importance': importances})
feature_importance_df

feature_importance_df = feature_importance_df.sort_values(by="Importance", ascending=False)

# Displaying feature importances
print("Feature Importances:")
print(feature_importance_df[:4])

# ANSWER ENDS HERE
```

Feature Importances:

	Feature	Importance
1	review_count	0.602031
2	checkins	0.130326
3	reviewer_stars	0.105590
0	business_stars	0.065582

1.1.8 Part F (14 Points Total)

Now you'll learn about clustering techniques! You'll perform data preprocessing, apply dimensionality reduction technique, and use K-means clustering to analyze the data. Finally, you'll visualize the results.

We wish to standardize the feature sets in order to make sure that the PCA is not biased by differences in scales within the data. Standardization aids with consistency in the data.

6.1.1 Normalize the data using StandardScaler. (1 pts)

- **Standardize** the features `X` and save it as `X_scaled`. (Include only the feature columns not the label column)

Hint: [documentation](#)

```
[59]: X = yelp_data[['business_id', 'business_stars', 'review_count', 'checkins', 'reviewer_stars', 'useful', 'funny', 'cool']]

# TODO: Normalize the data using StandardScaler
# ANSWER STARTING HERE
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X.drop(columns=['business_id']))

# ANSWER ENDS HERE
```

There are varying reasons for using a dimensionality reduction step such as PCA prior to data segmentation. By reducing the number of features, we are improving the performance of our algorithm. In addition, by decreasing the number of features the noise is also reduced.

PCA

Principal Component Analysis (PCA) is a dimensionality reduction technique that finds directions (principal components) maximizing variance in a dataset. It's an orthogonal linear transformation mapping data to a new coordinate system [1].

PCA steps for a dataset X with n samples and p features:

1. Center the data.
2. Compute covariance matrix $\Sigma = \frac{1}{n-1} X^T X$.
3. Calculate eigenvectors and eigenvalues of Σ .
4. Sort eigenvectors by descending eigenvalues.
5. Select top k eigenvectors as new basis.

Project original data onto this new basis for transformed data.

PCA is simple and effective for linear relationships but may not suit complex, non-linear structures [2].

For more in-depth reading on PCA:

[1] Scikit-learn PCA documentation: [sklearn.decomposition.PCA](#)

[2] Jolliffe, I. T. (2002). Principal Component Analysis, Second Edition. Springer Series in Statistics.

6.1.2 Use a PCA function to return the transformed data. (3 pts)

- Complete the PCA Implementation Function
- Apply Principal Component Analysis to Reduce Dimensionality
- Iteratively adjust the number of components `n_components` to retain at least 80% of the variance
- Save the resulting components for the elements in our data set as `X_pca`

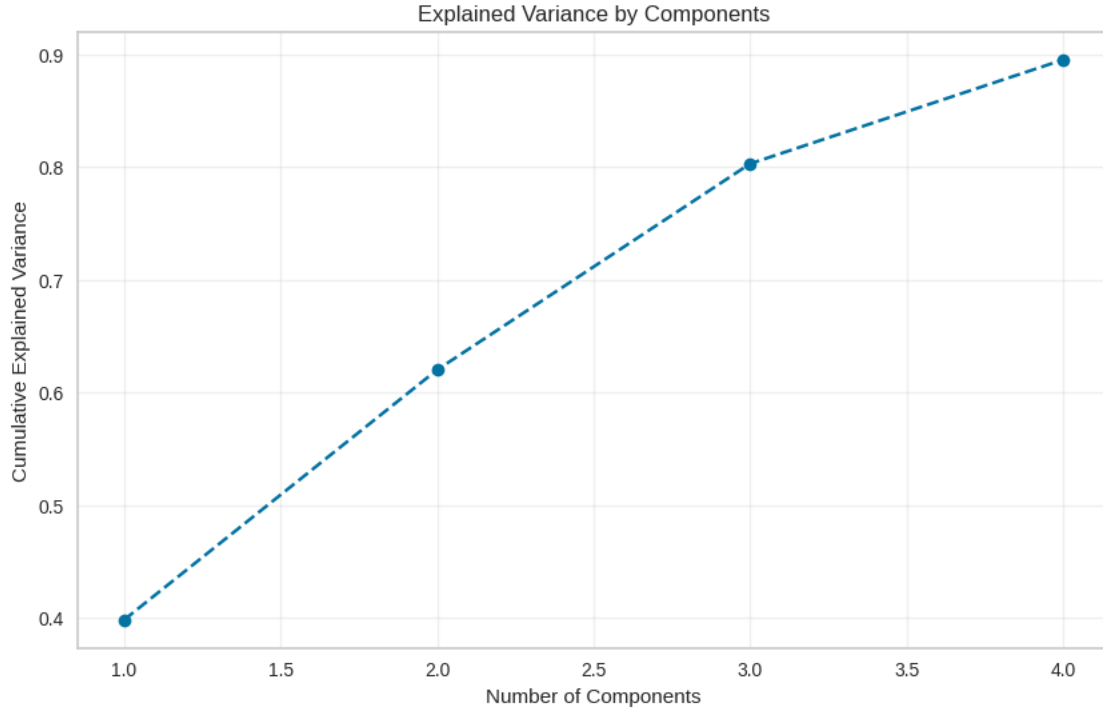
```
[60]: # TODO: Apply PCA and return the transformed data

def apply_pca(X_scaled, n_components):
    # ANSWER STARTING HERE

    pca = PCA(n_components=n_components)
    X_pca = pca.fit_transform(X_scaled)

    # ANSWER ENDS HERE
    plt.figure(figsize=(10, 6))
    plt.plot(range(1, n_components + 1), pca.explained_variance_ratio_.
    ↪cumsum(), marker='o', linestyle='--')
    plt.title('Explained Variance by Components')
    plt.xlabel('Number of Components')
    plt.ylabel('Cumulative Explained Variance')
    plt.grid(True, alpha=0.3)
    plt.show()

    return pca, X_pca
    # Apply PCA
n_components = 4
pca, X_pca = apply_pca(X_scaled, n_components)
print(X_pca.shape)
print(pca.explained_variance_ratio_)
```

(21318, 4)
[0.39838522 0.22179145 0.18285449 0.09222079]

6.2.1 Elbow Method (2.5 pts) In K-Means clustering, we start by randomly initializing k clusters and iteratively adjusting these clusters until they stabilize at an equilibrium point. However, before we can do this, we need to decide how many clusters (k) we should use.

In this part, you will determine the optimal number of clusters for a given dataset using the **Elbow Method**. The goal is to identify the point where adding more clusters results in minimal improvement in clustering quality, known as the “elbow point.”

The *distortion* for a given number of clusters k can be represented as:

$$D(k) = \frac{1}{N} \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where: - $D(k)$ is the distortion for k clusters - k is the number of clusters - C_i represents the i -th cluster - x is each data point in the cluster - μ_i is the centroid (mean) of the i -th cluster - $\|\cdot\|$ denotes the Euclidean distance

Task A: For each value of k from 1 to 9, first fit a **KMeans** model on `X_pca`. Then, using the resulting cluster assignments and centroids, **manually compute the mean distortion** using the formula above (**do not use `KMeans.inertia_`**)

- Store your mean distortion values in:

- A list named **distortions** (one value per k)
- A dictionary named **mapping** with keys as k values and values as their corresponding mean distortions
- Create an elbow plot with k on the x-axis and mean distortion on the y-axis
- Visually identify the elbow point (the optimal value of k)

Use `X_pca` that we calculated in the previous parts for this part. (**WARNING: You can't use the `KMeans.inertia_`**)

```
[61]: distortions = []
K = range(1, 10)
mapping = {}
for k in K:
    # ANSWER STARTING HERE

    kmeans= (KMeans(n_clusters=k, random_state=seed)).fit(X_pca)
    centers = kmeans.cluster_centers_
    labels = kmeans.labels_

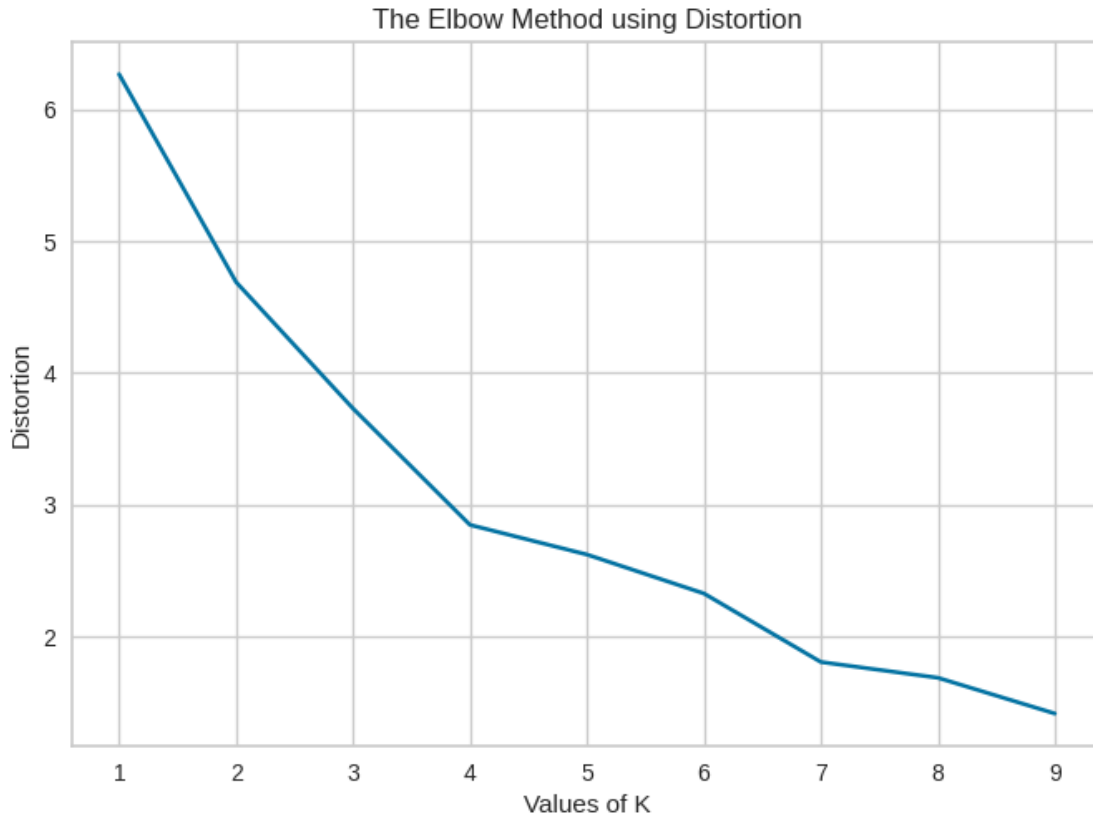
    sse = np.sum((X_pca - centers[labels])**2)
    mean_distortion = sse / X_pca.shape[0]
    distortions.append(mean_distortion)
    mapping[k] = mean_distortion

    # ANSWER ENDS HERE

for key, val in mapping.items():
    print(f'{key} : {val}')

plt.plot(K, distortions, 'bx-')
plt.xlabel('Values of K')
plt.ylabel('Distortion')
plt.title('The Elbow Method using Distortion')
plt.show()
```

```
1 : 6.266763553021727
2 : 4.689329334712707
3 : 3.729514429133743
4 : 2.846171049875379
5 : 2.621094597941368
6 : 2.3256477377887474
7 : 1.8054888277986156
8 : 1.6859572440797652
9 : 1.4145119312375862
```



6.2.2 KElbowVisualizer (2.5 pts) Task B: Use the `KElbowVisualizer` built-in function from the `yellowbrick` library to verify the conclusion above. See [Elbow Method](#), and [here](#) to install `yellowbrick`.

```
[62]: from sklearn import datasets
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
from yellowbrick.cluster import SilhouetteVisualizer
from yellowbrick.cluster import KElbowVisualizer
import yellowbrick.cluster.elbow as elbow_module
from yellowbrick.cluster.elbow import distortion_score

def mean_distortion_score(X, labels, metric="euclidean"):
    """Compute the mean distortion (average squared distance to centroids)"""
    total_distortion = distortion_score(X, labels, metric)
    return total_distortion / X.shape[0]

elbow_module.KELBOW_SCOREMAP['distortion'] = mean_distortion_score
```

```

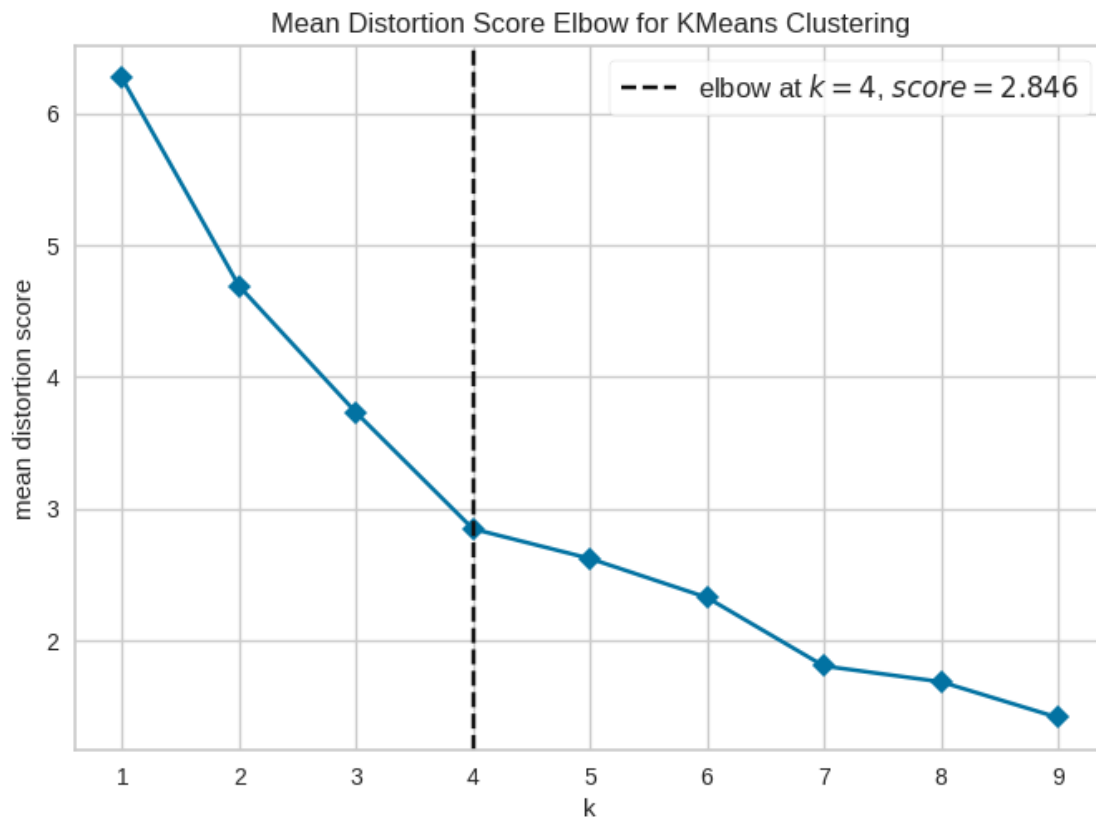
# ANSWER STARTING HERE
# Load the dataset
X_yellowbrick = X_pca

# Instantiate the clustering model and visualizer
model = KMeans(random_state=seed)

visualizer = KElbowVisualizer(model, k=(1, 10), metric='distortion',
    timings=False)

visualizer.fit(X_yellowbrick)
visualizer.show()
# ANSWER ENDS HERE

```



```

[62]: <Axes: title={'center': 'Mean Distortion Score Elbow for KMeans Clustering'},
      xlabel='k', ylabel='mean distortion score'>

```

To determine the optimal number of clusters, we have to select the value of k at the “elbow”, i.e. the point after which the distortion starts decreasing in a linear fashion. Thus for the given data, we conclude that the optimal number of clusters for the data is 4.

```
[63]: k_cluster = 4
```

6.2.3 Question: According to the plots above, why do you think we chose $K=4$? (1 pts)
Conceptually, what does the ‘distortion’ metric represent on the y-axis, and what does its decreasing trend indicate about the clustering?(1 pts) I think we chose $k=4$ because it represents the “elbow” of the plot. When you look at the graph, the line drops a good amount from $k = 1$ through $k = 4$. However, after $k = 4$, the line barely drops each time. The added complexity after this point is not worth the miniscule improvement in distortion. This follows the law of diminishing returns. Conceptually, distortion represents the lack of fit within the clusters. It measures how far away the restaurants are from the ideal middle of their cluster. High distortion means that the groups are spread out and messy, while low distortion means the groups are tighter. The decreasing trend indicates that as you add more clusters, the groups become more specific and tighter. Because the groups are tighter, there is less distance and thus less error between them.

K-Means Clustering

K-Means is a popular unsupervised learning algorithm used for clustering, where the goal is to partition data into K clusters based on feature similarity. The algorithm works by iteratively assigning data points to the nearest centroid (cluster center) and updating centroids to minimize the variance within each cluster. It starts by randomly initializing centroids and repeats the process until the centroids stabilize (convergence). K-Means is efficient for large datasets but sensitive to the choice of K and initial centroids. In your homework, you’ll implement K-Means to explore how well it groups your data based on similarity!

6.4 Apply K-means clustering and return the cluster labels (3 pts)

```
[64]: # TODO: Apply K-means clustering and return the cluster labels
def apply_kmeans(X, n_clusters, random_state=seed):
    # ANSWER STARTING HERE
    kmeans = KMeans(n_clusters=n_clusters, random_state=random_state)

    labels = kmeans.fit_predict(X)
    # ANSWER ENDS HERE
    return labels

# Apply K-means to both PCA results
pca_labels = apply_kmeans(X_pca, n_clusters = k_cluster)
pca_labels.shape
```

```
[64]: (21318,)
```

Visualize your clustering!

```
[65]: ## Step 6: Visualization

def visualize_3d_plotly_interactive(X_pca, pca_labels):
    """Fully interactive 3D visualization with Plotly using pre-computed
    labels"""
```

```

k_cluster = len(np.unique(pca_labels))

# Create the plot
fig = go.Figure()

# Add cluster points
colors = px.colors.qualitative.Set1[:k_cluster]
for i in range(k_cluster):
    cluster_mask = pca_labels == i
    fig.add_trace(go.Scatter3d(
        x=X_pca[cluster_mask, 0],
        y=X_pca[cluster_mask, 1],
        z=X_pca[cluster_mask, 2],
        mode='markers',
        marker=dict(
            size=5,
            color=colors[i],
            opacity=0.7
        ),
        name=f'Cluster {i}',
        text=[f'Cluster {i} Point {j}' for j in np.where(cluster_mask)[0]],
        hovertemplate='**%{text}**<br>' +
            'PC1: %{x:.2f}<br>' +
            'PC2: %{y:.2f}<br>' +
            'PC3: %{z:.2f}<br>' +
            '<extra></extra>'
    ))

# Calculate centroids from the labeled data
centroids = []
for i in range(k_cluster):
    cluster_points = X_pca[pca_labels == i]
    centroid = np.mean(cluster_points, axis=0)
    centroids.append(centroid)

centers_3d = np.array(centroids)[: , :3]

# Add centroids
fig.add_trace(go.Scatter3d(
    x=centers_3d[:, 0],
    y=centers_3d[:, 1],
    z=centers_3d[:, 2],
    mode='markers',
    marker=dict(
        size=8,
        color='red',
        symbol='x',
    )
))

```

```

        line=dict(width=2, color='darkred')
    ),
    name='Centroids',
    text=[f'Centroid {i}' for i in range(k_cluster)],
    hovertemplate='**%{text}**<br>' +
                  'PC1: %{x:.2f}<br>' +
                  'PC2: %{y:.2f}<br>' +
                  'PC3: %{z:.2f}<br>' +
                  '<extra></extra>'
))

# Update layout
fig.update_layout(
    title='Interactive 3D Cluster Visualization',
    scene=dict(
        xaxis_title='Principal Component 1',
        yaxis_title='Principal Component 2',
        zaxis_title='Principal Component 3',
        camera=dict(
            eye=dict(x=1.2, y=1.2, z=1.2)
        )
    ),
    width=900,
    height=700,
    showlegend=True
)

fig.show()

return pca_labels

# Usage with your existing pca_labels
cluster_labels = visualize_3d_plotly_interactive(X_pca, pca_labels)

```

You have been promoted!!

You are now one of the consulting company's elite employees!

1.1.9 Part G (2 points)

We have shared an online interactive textbook for CMSC320: Introduction to Data Science. The textbook is still in progress, and more chapters will be published gradually throughout the semester.

This short pre-feedback survey will help us understand your experience, preferences, and expectations regarding the textbook format before you start using it.

Here is the link to the survey: <https://forms.gle/n3Nbmeosoos7tjFa9>

Here is the link to the textbook: <https://ffalam.github.io/CMSC320TextBook/index.html>

Note: After completing the survey, please take a screenshot of your submission and submit it as

proof of participation. This is a compulsory question. If you have already completed the survey earlier, you do not need to do it again, but you must submit the previous screenshot to receive the points.

1.1.10 Extra Credit (5 points total)

The matplotlib and seaborn libraries are great tools for visualizing data.

They are used to create legible and complex graphs that are essential in data interpretation and analysis.

7.1 Visualize the distribution of each feature in the original dataset for different rating classes (2 pts)

- **Plot** out histograms for each feature. You can use a for loop.
- Label the **title** of each histogram “Distribution of (Feature Name) by Rating”
- Label the **axes** appropriately and display each histogram.
- Display the **y-axis** using a **logarithmic** scale.

```
[66]: histogram_pd = X
      histogram_pd.head()

      rating_classes = yelp_data['preference_rating']

      # Create subplots
      n_features = histogram_pd.shape[1] - 1
      n_cols = 3
      n_rows = int(np.ceil(n_features / n_cols))

      fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, 12))
      axes = axes.flatten() # Make it easier to index

      # ANSWER STARTING HERE

      # Plot histogram for each feature
      for i, feature in enumerate(histogram_pd.columns[:-1]):
          ax = axes[i]

          # Get unique rating classes
          unique_ratings = rating_classes.unique()

          # Plot histogram for each rating class
          for rating in unique_ratings:
              data = histogram_pd[rating_classes == rating][feature]

              ax.hist(data, bins=20, alpha=0.5, label=f'Rating {rating}')
```

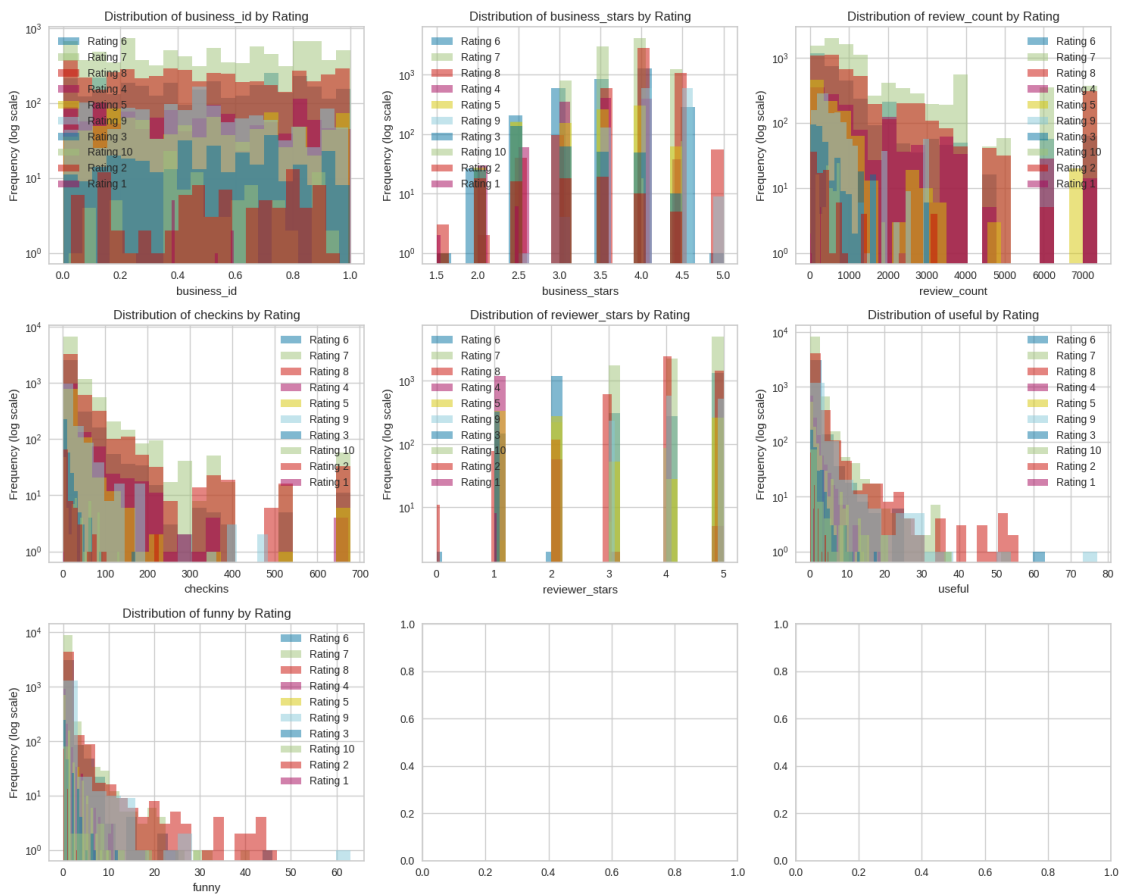


```
# Set logarithmic scale for y-axis
ax.set_yscale('log')

# Set labels and title
ax.set_title(f'Distribution of {feature} by Rating')
ax.set_xlabel(feature)
ax.set_ylabel('Frequency (log scale)')
ax.legend()
```

ANSWER ENDS HERE

```
plt.tight_layout()
plt.show()
```



7.2 Explore correlations between features and Rating ratings (1 pts)

- Create a **correlation matrix** based on the correlations between features and rating.
- **Display** the matrix.

```
[67]: # Correlation Matrix

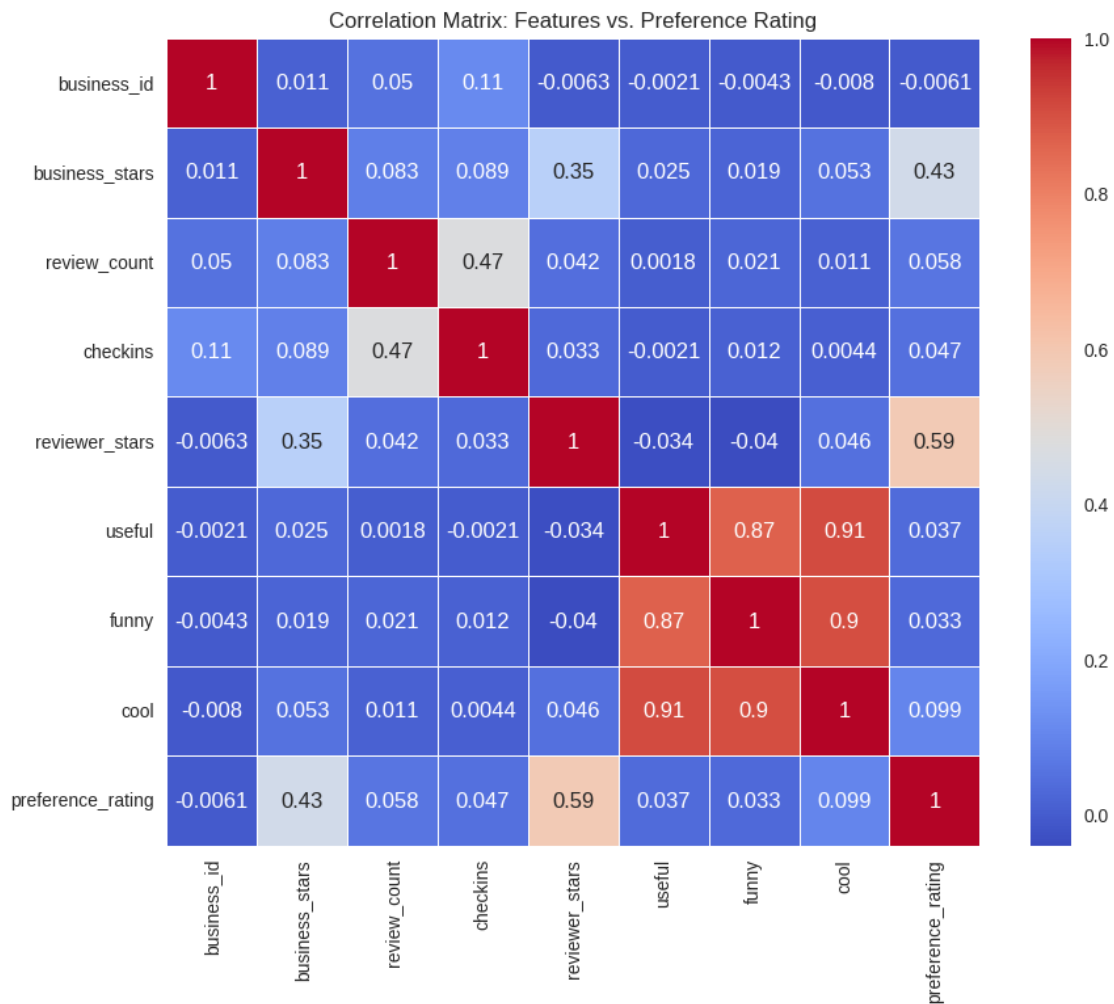
# ANSWER STARTING HERE
correlation_data = X.copy()
correlation_data['preference_rating'] = yelp_data['preference_rating']

correlation_matrix = correlation_data.corr()

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths = 0.5)
plt.title('Correlation Matrix: Features vs. Preference Rating')
plt.show()

print("Correlation with Preference Rating:")
print(correlation_matrix['preference_rating'].sort_values(ascending=False))

# ANSWER ENDS HERE
```



Correlation with Preference Rating:

preference_rating	1.000000
reviewer_stars	0.586658
business_stars	0.434996
cool	0.098603
review_count	0.057882
checkins	0.046575
useful	0.036551
funny	0.032535
business_id	-0.006077

Name: preference_rating, dtype: float64

7.3 Question: What have you learned from this correlation map? Why? (2 pts) The most significant takeaway from the correlation map is that reviewer_stars and business_stars are the only decently strong predictors of preference_rating. We see that cool, useful, funny, and checkins have almost 0 correlation to preference_rating. This makes sense because a funny review might actually just be a 1-star rant. Review_count and checkins measure popularity, not quality, so it makes sense why it doesn't really contribute to a "top tier" restaurant.